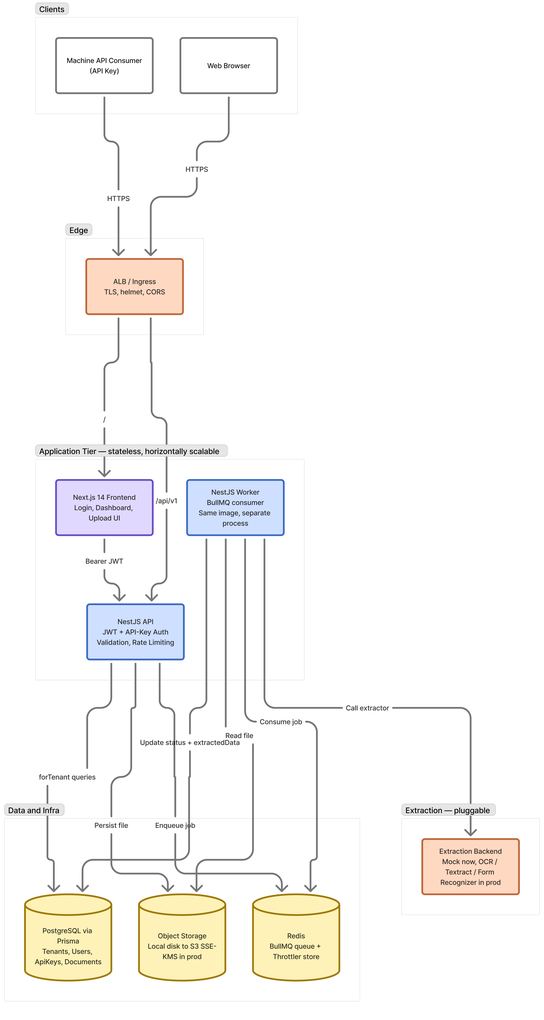
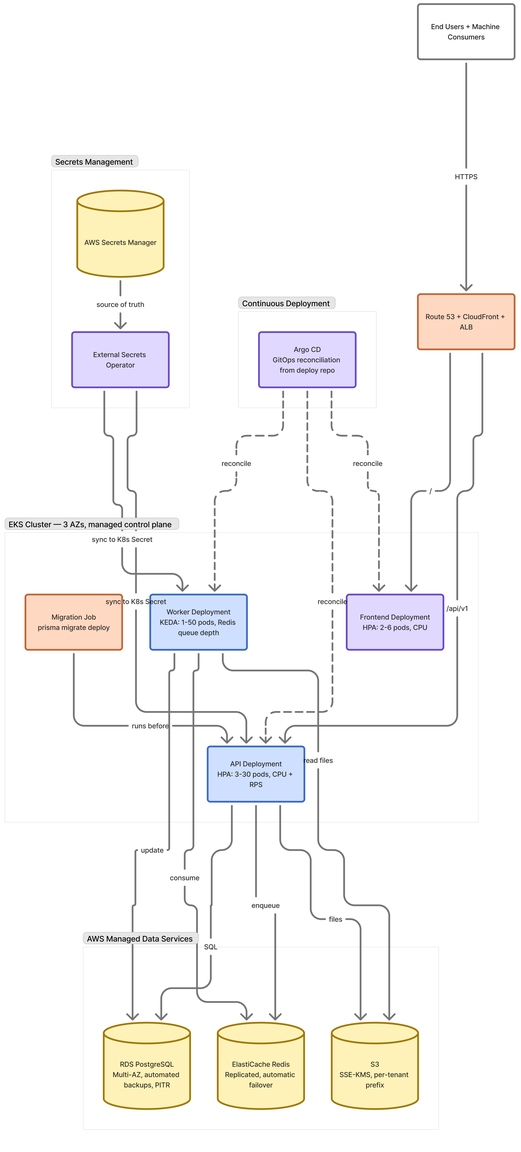
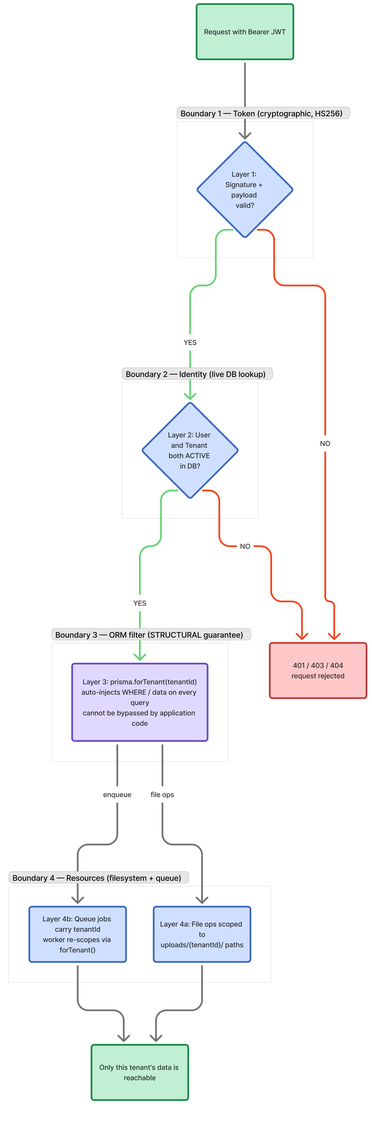
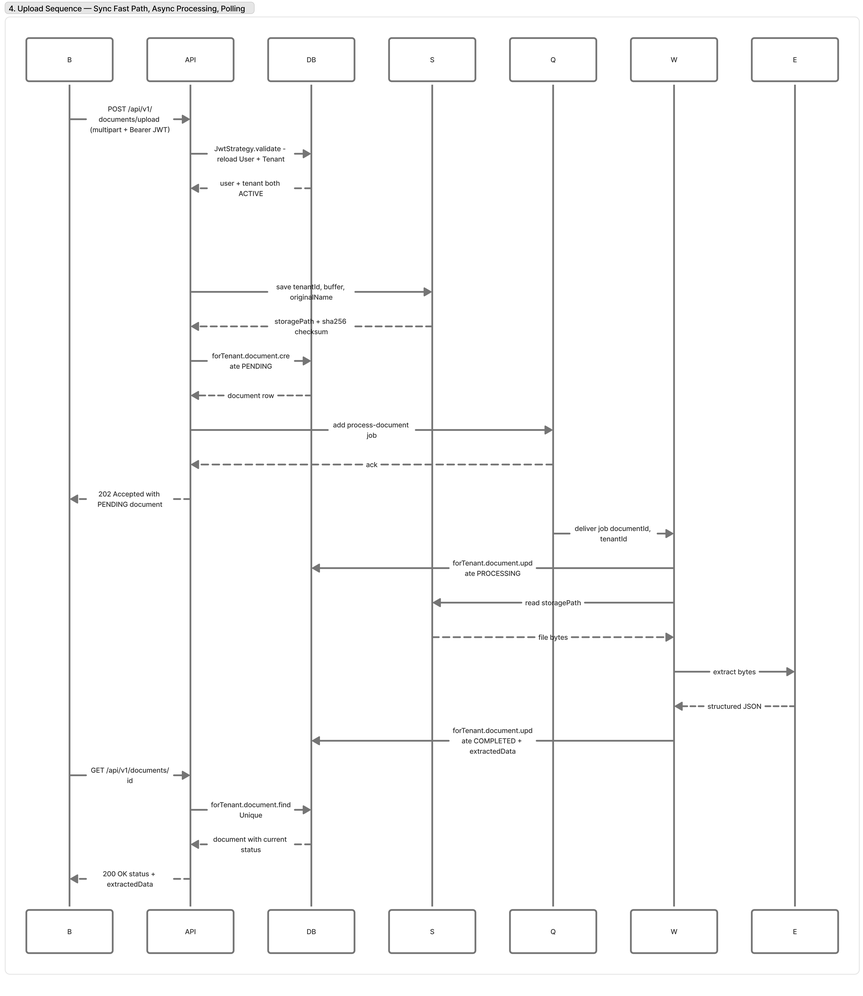

Clients
Machine API Consumer (API Key)
Web Browser
HTTPS
HTTPS
Edge
ALB / Ingress TLS, helmet, CORS
/
Application Tier — stateless, horizontally scalable
Next.js 14 Frontend Login, Dashboard, Upload UI
/api/v1
NestJS Worker BullMQ consumer Same image, separate process
Bearer JWT
NestJS API JWT + API-Key Auth Validation, Rate Limiting
Call extractor
Consume job
Read file
Update status + extractedData
forTenant queries
Persist file
Enqueue job
Data and Infra
PostgreSQL via Prisma Tenants, Users, ApiKeys, Documents
Object Storage Local disk to S3 SSE-KMS in prod
Redis BullMQ queue + Throttler store
Extraction — pluggable
Extraction Backend Mock now, OCR / Textract / Form Recognizer in prod


End Users + Machine Consumers
HTTPS
Route 53 + CloudFront + ALB
/
Secrets Management
AWS Secrets Manager
source of truth
External Secrets Operator
Continuous Deployment
Argo CD GitOps reconcillation from deploy repo
reconcile
reconcile
reconcile
EKS Cluster — 3 AZs, managed control plane
sync to K8s Secret
sync to K8s Secret
Migration Job prisma migrate deploy
Worker Deployment KEDA: 1-50 pods, Redis queue depth
Frontend Deployment HPA: 2-6 pods, CPU
/api/v1
runs before
API Deployment HPA: 3-30 pods, CPU + RPS
read files
update
consume
enqueue
files
SQL
AWS Managed Data Services
RDS PostgreSQL Multi-AZ, automated backups, PITR
ElastiCache Redis Replicated, automatic failover
S3 SSE-KMS, per-tenant prefix


Request with Bearer JWT
Boundary 1 — Token (cryptographic, HS256)
Layer 1: Signature + payload valid?
YES
NO
Boundary 2 — Identity (live DB lookup)
Layer 2: User and Tenant both ACTIVE in DB?
YES
NO
401 / 403 / 404 request rejected
Boundary 3 — ORM filter (STRUCTURAL guarantee)
Layer 3: prisma.forTenant(tenantId) auto-injects WHERE / data on every query cannot be bypassed by application code
enqueue
file ops
Boundary 4 — Resources (filesystem + queue)
Layer 4b: Queue jobs carry tenantId worker re-scopes via forTenant()
Layer 4a: File ops scoped to uploads/{tenantId}/ paths
Only this tenant's data is reachable


4. Upload Sequence — Sync Fast Path, Async Processing, Polling
B
API
DB
S
Q
W
E
POST /api/v1/ documents/upload (multipart + Bearer JWT)
JwtStrategy.validate - reload User + Tenant
user + tenant both ACTIVE
save tenantId, buffer, originalName
storagePath + sha256 checksum
forTenant.document.cre ate PENDING
document row
add process-document job
ack
202 Accepted with PENDING document
deliver job documentId, tenantId
forTenant.document.upd ate PROCESSING
read storagePath
file bytes
extract bytes
structured JSON
forTenant.document.upd ate COMPLETED + extractedData
GET /api/v1/documents/ id
forTenant.document.find Unique
document with current status
200 OK status + extractedData